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Editorial Office

Information Sciences

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Dear Editor,

We are pleased to submit our manuscript, "Structurally-Calibrated Functional Metacognitive Attribution for Audit Prioritization in Knowledge-Intensive Reasoning", for consideration as a regular article in Information Sciences.

The manuscript fits Information Sciences because it addresses a core information and knowledge representation problem: how exposed intermediate knowledge artifacts can be represented, organized, inspected, and reused when human curation budgets are limited. Modern knowledge-intensive systems expose retrieval checks, entity bindings, graph nodes, verification records, process annotations, and reasoning traces, but these artifacts are often available only as heterogeneous signals rather than reusable audit records.

We introduce SC-FMA as a knowledge representation and transformation layer for this setting. SC-FMA converts observable intermediate artifacts into structured audit records that track supplied annotation or utility fidelity while adding graph-aware structural-dependency, redundancy, bottleneck, audit-reason, and interpretation fields. The novelty lies in a reusable audit-record schema that turns intermediate artifacts into maintainable knowledge objects.

The contribution to information and knowledge processing is representational. The resulting records support fixed-budget audit-oriented analysis, knowledge curation, redundancy interpretation, dependency inspection, weak-signal diagnosis, and reuse of curated artifacts across the knowledge lifecycle. The manuscript makes this contribution concrete through four audit categories: weak utility anchor, structural over-correction, redundancy consolidation, and bottleneck protection. It studies these categories through a locked process-annotation dataset, a fixture-level Countries-KG backend feasibility study, rule-derived audit-record construction, and controlled calibration analysis.

The manuscript presents SC-FMA as a knowledge representation layer and explicitly separates the scalar view, raw-field bundle, and SCU-calibrated audit record. This clarifies where field exposure alone is sufficient and where SCU calibration adds audit organization beyond simpler views. We believe the submission is well-aligned with the scope of Information Sciences.

The reported evidence is limited to observable intermediate artifacts and fixed-budget audit settings, and the manuscript treats this boundary as part of its methodological discipline. The strongest supervised fidelity field remains the best PRM800K ranking signal, while SC-FMA contributes decomposed audit records and category-level organization rather than claiming an independent ranking gain. A same-supervision structure-only control and direct graph-necessity diagnostics show that position and lexical cues, rather than graph topology alone, carry most of the recoverable PRM800K annotation-order signal. A windowed QP analysis is reported only as post hoc diagnosis of the observed long-trace failure. The Countries-KG study is a graph-backend feasibility and methodological-analogy check, not production knowledge-base validation. No human-rater experiment is included; human audit-use studies and external task generalization require separate evaluation.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere.

This work was supported by the National Social Science Fund of China Project (24BSH018) and the Beijing Natural Science Foundation Project (L252145). The authors declare that they have no conflicts of interest related to this work. The manuscript's generative-AI disclosure states that OpenAI GPT-5 was used to improve language clarity and assist with LaTeX formatting checks, not to design the study, conduct data analysis, generate results, or derive conclusions, with all content reviewed by the authors. The PRM800K process-annotation dataset, MuSiQue, and WebQSP are publicly available from their original sources; derived reports and reproduction artifacts will be made available through an anonymous review repository during submission and released with the final article where permitted.

Correspondence may be directed to Ningning Wang at wangningning@bistu.edu.cn. Haoran Ma can be reached at mahaoran0000@foxmail.com.

Sincerely,

Haoran Ma and Ningning Wang